Shukla Ashish Anand

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Education

Indian Institute of Science Education and Research (IISER), Bhopal — Bhopal, MP

Bachelor of Technology (B.Tech) in Data Science and Engineering, ***CPI: 8.22 / 10.0*** — *Aug. 2024 – May 2028 (Expected)*

- **Relevant Coursework**: Data Structures & Algorithms, Probability & Statistics, Linear Algebra, Discrete Mathematics, Econometrics, Signals & Systems, Calculus, Multivariable Calculus, Programming.

Technical Skills

Research Interests: Robust & Efficient ML, Multimodal Learning, Speech & Audio Processing, Representation Learning, Domain Adaptation, Test-Time Adaptation.
Languages: Python, C/C++.
ML Frameworks: PyTorch, Hugging Face Transformers, PEFT, LoRA, QLoRA, bitsandbytes.
Speech & Vision: Whisper, Qwen3-ASR, torchaudio, librosa, ASR, VSR, Audio-Visual Speech Recognition (AVSR), OpenCV, VideoMAE.
Tools: Git, Linux/Bash, Weights & Biases, Remote GPU Clusters, NumPy, Pandas.
Core Competencies: Machine Learning, Computer Vision, State Space Models (Mamba), Model Evaluation, Probability & Statistics.

Experience

Speech Lab, SCSE @ Nanyang Technological University (NTU) — Singapore

Research Intern – Speech Data & ASR Evaluation — *Aug. 2026 – Dec. 2026*

- Working on **speech data quality, accent robustness, and Automatic Speech Recognition (ASR)**, studying how data curation and language identification affect the reliability and representativeness of speech datasets.
- Evaluating **Whisper and Qwen3-ASR** on accented English, investigating the effects of data filtering, text normalization, decoding strategies, and parameter-efficient adaptation on recognition performance.

Speech Lab, LTRC @ IIIT Hyderabad — Hyderabad, India

Research Intern – Speech & Audio Modeling — *May 2026 – Jul 2026*

- Developed an information-theoretic pitch stylization pipeline using Shannon Entropy, Mutual Information, and KL Divergence for robust speech representation modeling.
- Applied Dynamic Programming and Weighted Least Squares regression for optimal piecewise pitch contour segmentation.
- Derived a Bayesian Information Criterion (BIC/MDL) for automatic model-complexity selection, achieving a **38% RMSE reduction** over heuristic baselines.

Visual Data Computing Group (VisDom) — Bhopal, India

Research Intern – Multimodal Machine Learning — *2025 – May 2026*

- Worked under Prof. Vinod K. Kurmi on robust multimodal and speech modeling, contributing to **PRISM, Mamba-AVSR, and VaaniAdapt**.

Publications & Projects

PRISM (First Author) | *Accepted at ACM CIKM 2026, Rome, Italy* — 2026

- **Prototype-Rectified Iterative Self-supervised Manifold Denoising under Severe Acoustic Shift.**
- Developed a **training-free, source-free test-time adaptation** framework for audio-text foundation models under severe acoustic domain shifts.
- Modeled acoustic noise as a low-rank affine distortion in multimodal embedding space and reversed it using **frozen text prototypes as geometric anchors**, requiring no gradient updates.
- Designed a closed-form adaptation pipeline for efficient inference under previously unseen acoustic conditions; paper: arxiv.org/abs/2608.15037.

Pitch Stylization | *Under Review at ICASSP 2027* — 2026

- **Mutual-Information-Guided Multi-Estimator Pitch Fusion for Robust Pitch Stylization.**
- Developed a mutual-information-guided multi-estimator framework that combines complementary pitch estimators for robust pitch stylization.

Mamba-AVSR | *PyTorch, Mamba-2, Whisper-v3, VideoMAE, Llama-3.2* — Dec. 2025 – Present

- Prototyped an end-to-end Audio-Visual Speech Recognition pipeline integrating Whisper-v3 and VideoMAE encoders with a Llama-3.2 backend.
- Replaced quadratic $O(N^2)$ cross-attention with a linear-time $O(N)$ Mamba-2 connector to efficiently fuse asynchronous audio and video streams.

- Explored preference optimization to improve transcription reliability under challenging acoustic conditions.

VaaniAdapt | *Continual Learning, Multilingual ASR* 2026

- Proposed a rehearsal-free continual learning framework for adapting deployed speech recognition systems to new languages without storing or replaying prior speech data.
- Designed low-rank residual adapters with orthogonality constraints to reduce interference between language-specific adaptations.

DermaScribe | *System Architect & AI Lead, Python, SQLite, OpenCV* Nov. 2025

- Architected a hybrid offline-online healthcare database using SQLite and FTS5 indexing, achieving sub-millisecond search for structured queries.
- Developed a local caching layer for AI API responses to reduce inference cost and latency for repetitive queries.
- Built an OpenCV-based facial skin-condition screening module and integrated the backend into a Streamlit interface.
- Awarded **1st Place (Healthcare Track)** at the IISER Bhopal DSE Day Hackathon.